

Engineering framework for curiosity-driven and humble AI in clinical decision support

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ABSTRACT

We present BODHI (Balanced, Open-minded, Diagnostic, Humble, and Inquisitive), an engineering framework for curiosity driven and humble clinical decision support artificial intelligence (AI) systems. Despite growing capabilities, large language models (LLMs) often express inappropriate confidence, conflating statistical pattern recognition with genuine medical understanding. BODHI addresses this through a dual reflective architecture that: (1) decomposes epistemic uncertainty into task specific dimensions, and (2) constrains model responses using virtue based stance rules derived from a Virtue Activation Matrix. We validate the framework through controlled evaluation on 200 clinical vignettes from HealthBench Hard, assessing GPT-4o-mini and GPT-4.1-mini across 5 random seeds (2000 total observations). Statistical analysis included bootstrap resampling, paired t tests, and effect size computation. BODHI improved overall clinical response quality (GPT-4.1-mini: +16.6 pp, $p<0.0001$, Cohen's $d=11.56$; GPT-4o-mini: +2.2 pp, $p<0.0001$, Cohen's $d=1.56$) and achieved very large effect sizes on curiosity (context seeking rate: Cohen's $d=16.38$ and 19.54) and humility (hedging: $d=5.80$ for GPT-4.1-mini) metrics. Crucially, 97.3% of GPT-4.1-mini responses and 73.5% of GPT-4o-mini responses included appropriate clarifying questions, compared with 7.8% and 0.0% at baseline, demonstrating the framework's effectiveness in eliciting information gathering behaviour. Findings suggest LLMs can be reliably constrained to operate within epistemic boundaries when provided with structured uncertainty decomposition and virtue aligned response rules, offering a pathway towards safer clinical AI deployment

INTRODUCTION

Medical errors remain a leading cause of death, claiming over 250 000 lives annually in the USA.¹ Artificial intelligence (AI) systems that express unwarranted confidence can contribute to such errors in clinical settings: experienced radiologists might follow incorrect AI suggestions despite contradictory evidence,² while critical care clinicians

defer to systems even when clinical intuition suggests otherwise.^{3 4} This often happens when presented with high confidence scores.²

Large language models (LLMs) regularly exhibit overconfidence in clinical reasoning tasks. Recent benchmarking evaluations show that even the most accurate LLMs demonstrate minimal variation in confidence between correct and incorrect answers, limiting their safe use in clinical settings.⁵ LLMs show inflexible reasoning and propensity to hallucinate when faced with scenarios requiring deviation from training patterns⁶ and exhibit sycophantic behaviour (excessively praising or flattering a person in authority, in this case the prompter/user) with up to 100% compliance, even when presented with illogical medical requests.⁷ Uncertainty estimation analyses indicate overconfidence despite limited accuracy,^{6 7} with confidence scores failing to align with prediction accuracy across models and medical specialties.⁸

Several approaches have attempted to address AI overconfidence. Uncertainty quantification methods and conformal prediction can estimate epistemic uncertainty.⁹⁻¹¹ However, these approaches focus on measuring uncertainty rather than modifying behaviour when uncertain. Lindenmeyer *et al*⁹ compared ensemble and Gaussian process approaches for uncertainty estimation in clinical decision support, yet systems still produced confident outputs regardless of estimated uncertainty. Chain-of-thought prompting has improved medical reasoning accuracy and interpretability,¹²⁻¹⁴ but existing implementations optimise for correctness rather than appropriate epistemic conduct. Fine-tuning approaches like ConfiDx¹⁵ show promise for uncertainty-aware diagnosis but

require model modification and focus narrowly on diagnostic tasks.

Previous work has called for epistemic humility as a design principle,¹⁶ arguing epistemic humility in AI-assisted pain assessment requires ethics education for developers and clinicians, but stops short of engineering implementations. Madhukar¹⁷ formalises the ‘AI Socratic Paradox’ where clinical AI must know when to deviate from its own recommendations, highlighting the need for metacognitive self-awareness but not providing concrete solutions. Uncertainty is often quantified without behavioural modification and epistemic virtues are discussed without operationalisation, or they require model fine-tuning that limits practical deployment. No prompting-based framework has demonstrated empirical effectiveness in constraining LLM behaviour to align with epistemic virtues in clinical contexts.

We previously introduced curiosity and humility as essential epistemic virtues for healthcare AI.^{18–20} Curiosity represents the drive to reduce uncertainty through targeted inquiry, while humility involves acknowledging limitations and deferring appropriately to human expertise. Building on this conceptual foundation, the present work addresses the critical gap between philosophical proposals and engineering implementation. Our framework for balanced, open-minded, diagnostic, humble and inquisitive machines (BODHI) fills this gap through a two-pass chain-of-thought protocol. The first pass generates structured uncertainty analysis; the second produces appropriately constrained responses guided by a Virtue Activation Matrix. We validate BODHI on 200 clinical vignettes from HealthBench Hard,²¹ demonstrating improvements in context-seeking behaviour.

METHODS

Framework architecture

The BODHI framework operates through six integrated steps (online supplemental materials figure S1). First, clinical complexity assessment evaluates the incoming query across multiple dimensions including diagnostic ambiguity, urgency indicators and data completeness. Second, prior confidence evaluation estimates the model’s epistemic state based on training distribution coverage and query specificity. Third, parallel Curiosity and Humility modules process the clinical scenario: the Curiosity module identifies information gaps and generates clarifying questions, while the Humility module assesses confidence boundaries and deferral triggers. Fourth, the Virtue Activation Matrix maps the combined outputs to one of four epistemic stances. Fifth, adaptive system responses are generated according to the selected stance. Sixth, continuous learning mechanisms incorporate clinician feedback to refine threshold calibrations over time.

Virtue Activation Matrix

The Virtue Activation Matrix (figure 1) maps clinical complexity against model confidence to determine appropriate epistemic stances. Four quadrants organise behavioural responses based on the interplay between uncertainty and stakes.

Quadrant 1: Proceed and Monitor applies when confidence is high and complexity is low, requiring minimal curiosity and low humility burden.

Quadrant 2: Watchful and Alternatives activates under high confidence but high complexity, demanding situational awareness and consideration of differential diagnoses.

Quadrant 3: Clarify and Review governs low confidence with low complexity scenarios, triggering high curiosity and epistemic humility through targeted clarifying questions.

Quadrant 4: Escalate and Reframe represents the most demanding stance: low confidence under high complexity requires high curiosity combined with deferential humility, explicitly escalating to human expertise.

The decision logic for stance selection follows a tree of conditional probabilities based on clinical complexity indicators (online supplemental material figure S2). Operational thresholds for confidence and complexity classification are detailed in online supplemental material S5.1.

Two-Pass Chain-of-Thought Protocol

The BODHI protocol operationalises epistemic virtues through structured chain-of-thought scaffolding that separates internal reasoning from external communication.

Pass 1 generates structured analysis with seven mandatory fields (online supplemental table S1): task type classification (Emergency, Technical, Hybrid, or Conversation), audience identification (Patient, Health Professional or Unclear), primary hypothesis with reasoning, key uncertainties affecting confidence, clarifying questions (1–2 required for non-emergency cases), red flags triggering escalation, and safe recommendations appropriate to the uncertainty level. This structured output makes the model’s epistemic state explicit and auditable.

Pass 2 produces the clinician-facing response by conditioning on pass 1 analysis and applying epistemic constraints (online supplemental table S2). Task routing rules adjust behaviour based on context: Emergency Mode prioritises safety over completeness, Technical Mode reduces hedging for administrative tasks, Hybrid Mode balances clinical reasoning with technical precision and Conversation Mode (default) applies full epistemic constraints for patient-facing interactions. Cross-cutting constraints enforce specificity (concrete numbers and timeframes when available), active inquiry (converting conditional statements into direct questions) and presentation of alternatives (acknowledging differential diagnoses when confidence is below threshold). Complete protocol specifications including the full prompting

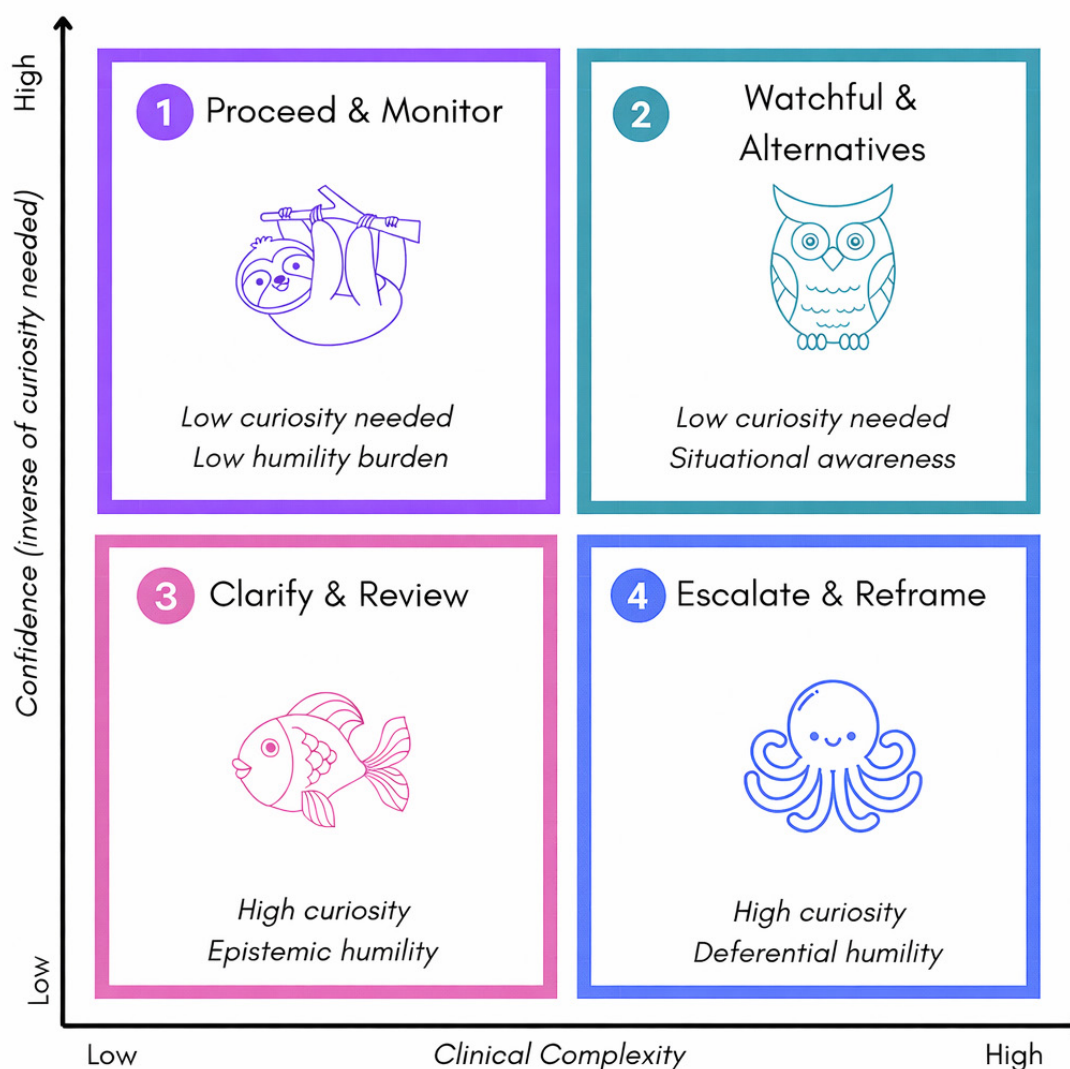


Figure 1 Virtue Activation Matrix mapping clinical complexity against model confidence. Each quadrant represents a distinct epistemic stance with associated curiosity and humility requirements. Q1 (Proceed and Monitor): routine cases with high confidence requiring low curiosity and low humility burden. Q2 (Watchful and Alternatives): complex cases requiring situational awareness. Q3 (Clarify and Review): uncertain cases needing high curiosity and epistemic humility. Q4 (Escalate and Reframe): high-stakes uncertainty demanding high curiosity and deferential humility with explicit escalation to human expertise.

templates are provided in online supplemental material S2.

Evaluation Design

We evaluated BODHI using HealthBench Hard,²¹ a benchmark of 200 challenging clinical scenarios requiring diagnostic reasoning, treatment planning and triage decisions. HealthBench Hard represents a curated subset selected for cases requiring nuanced clinical judgement where baseline models show suboptimal performance. The dataset spans multiple clinical domains, including emergency medicine, primary care and specialty consultations.

Two frontier language models were assessed: GPT-4.1-mini and GPT-4o-mini across 5 random seeds (42, 43, 44, 45, 46) yielding 1000 case-level observations. Each model was evaluated under baseline (standard prompting) and BODHI (two-pass protocol) conditions. This seed-based

replication design enables quantification of reproducibility and controls for sampling variance. Model configuration details including temperature (0.7), token limits (4096) and API parameters are provided in online supplemental material S3.2), online supplemental table S3.

Statistical analysis employed non-parametric bootstrap resampling (10 000 iterations) for CIs, case-level paired t-tests accounting for repeated measures structure, mixed-effects models with random intercepts for seed and case, and Cohen's d with Hedge's g correction for effect sizes (online supplemental material S3.4). Primary outcomes included overall HealthBench score (composite rubric aggregating accuracy, safety and communication), context-seeking rate (proportion of responses containing at least one appropriate clarifying question), hedging frequency (explicit uncertainty acknowledgement),

Table 1 Primary outcomes: baseline versus BODHI performance

Metric	Baseline	Bodhi	Δ (95% CI)	Cohen's d
GPT-4.1-mini (n=1000, 5 seeds)				
Overall score	2.5%±1.8%	19.1%±1.0%	+16.6 pp (15.5 to 17.7)	11.56
Context-seeking rate	7.8%±6.8%	97.3%±3.7%	+89.6 pp (81.4 to 96.2)	16.38
Hedging	1.7%±1.7%	21.9%±4.6%	+20.3 pp (17.1 to 23.4)	5.80
Communication quality	70.1%±5.4%	57.5%±2.6%	-12.5 pp (-14.8 to -10.1)	-2.94
GPT-4o-mini (n=1000, 5 seeds)				
Overall score	0.0%±0.0%	2.2%±2.0%	+2.2 pp (0.8 to 4.0)	1.56
Context-seeking rate	0.0%±0.0%	73.5%±5.3%	+73.5 pp (69.6 to 77.7)	19.54
Hedging	0.0%±0.0%	4.1%±5.0%	+4.1 pp	1.16
Communication quality	62.7%±4.6%	51.4%±6.5%	-11.3 pp (-15.9 to -6.8)	-2.02

Values shown as mean±SD across five seeds. Effect size interpretation: small (0.2), medium (0.5), large (0.8), very large (>1.2). 95% CIs computed via bootstrap resampling (10000 iterations).
BODHI, Balanced, Open-minded, Diagnostic, Humble and Inquisitive; pp, percentage points.

helpful and safe score, emergency referral rate and communication quality. Complete metric definitions are provided in online supplemental material S4.1 and online supplemental table S4.

RESULTS

Primary Outcomes

BODHI produced significant improvements across both models (table 1). For GPT-4.1-mini, overall score improved from 2.5% to 19.1% (+16.6 percentage points, 95% CI (15.5 to 17.7), $p<0.0001$). Context-seeking rate increased from 7.8% to 97.3% (+89.6 pp), representing a very large effect (Cohen's $d=16.38$). Hedging behaviour increased from 1.7% to 21.9% (+20.3 pp, 95% CI (17.1 to 23.4), $d=5.80$). For GPT-4o-mini, overall score improved from 0.0% to 2.2% (+2.2 pp, 95% CI (0.8 to 4.0), $p<0.0001$), with context-seeking rate increasing from 0.0% to 73.5% (+73.5 pp, 95% CI (69.6 to 77.7), $d=19.54$). These effect sizes substantially exceed conventional thresholds for

large effects ($d>0.8$), indicating robust behavioural modification rather than marginal improvement.

Multimetric performance comparison

Figure 2 presents comprehensive multimetric comparisons for both models. For GPT-4.1-mini (figure 2, left panel), BODHI produced significant improvements across five of six metrics: Overall Score, Context-Seeking Rate, Helpful and Safe, Emergency Referrals and Hedging. The only metric showing a decrease was Communication Quality, reflecting the expected trade-off when epistemic constraints convert confident declarations into appropriately hedged, question-containing responses. For GPT-4o-mini (figure 2), right panel), the pattern was consistent with GPT-4.1-mini. BODHI enhanced curiosity as measured by context-seeking rate, humility as measured by hedging and safe recommendations, and safety as measured by emergency referrals. The magnitude of improvements differed between models, with GPT-4.1-mini showing larger gains in overall score

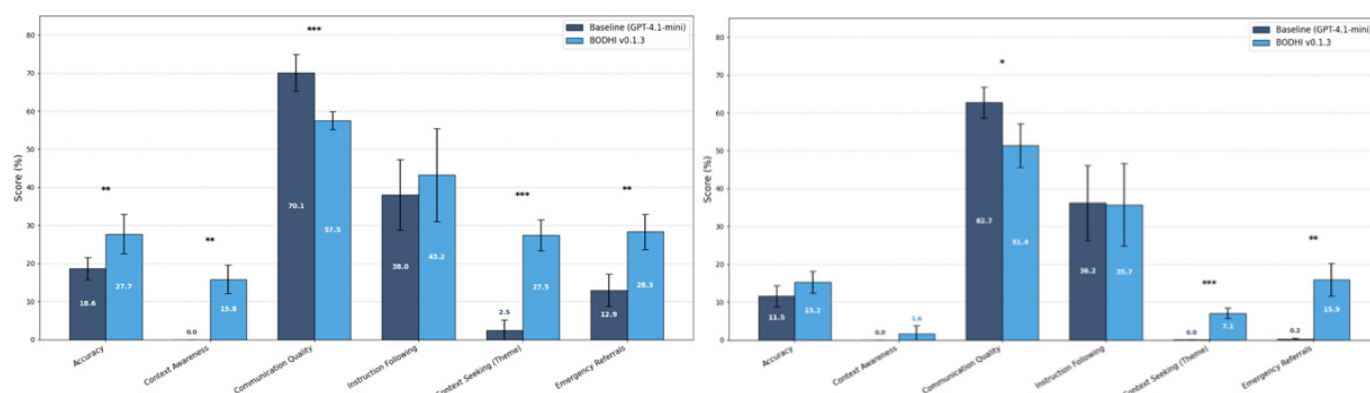


Figure 2 Multimetric comparison for GPT-4.1-mini (left) and GPT-4o-mini (right) across five random seeds (n=1000 cases per model). Error bars represent 95% CIs. BODHI framework (green) versus baseline (blue) showing significant improvements in Overall Score, Context-Seeking Rate, Helpful and Safe, Emergency Referrals and Hedging, with expected decreases in Communication Quality reflecting epistemic constraint adherence. * $p<0.05$, ** $p<0.01$, *** $p<0.001$. BODHI, Balanced, Open-minded, Diagnostic, Humble and Inquisitive.

and hedging behaviour. Both models demonstrated modest communication quality trade-offs, confirming that the epistemic constraint mechanism operates consistently across model variants. Error bars representing 95% CIs demonstrate tight clustering around mean estimates, indicating stable performance across the 1000 case-level observations per model.

Effect Size Analysis

To assess the practical significance of BODHI's effects beyond statistical significance, we computed standardised effect sizes (Cohen's *d*) for all primary metrics (online supplemental figure S3). Both models demonstrated very large effects on curiosity metrics, with context-seeking showing $d=16.38$ for GPT-4.1-mini and $d=19.54$ for GPT-4o-mini. These effect sizes are substantially larger than conventional thresholds: small (0.2), medium (0.5) and large (0.8). The exceptionally large effects on context-seeking reflect the near-ceiling performance achieved by BODHI (97.3% and 73.5%) compared with near-floor baseline performance (7.8% and 0.0%).

GPT-4.1-mini demonstrated stronger hedging responses ($d=5.80$) compared with GPT-4o-mini ($d=1.16$), suggesting that model capacity influences the effectiveness of humility constraint application.

Overall score improvements showed very large effects for both models ($d=11.56$ and $d=1.56$, respectively). The negative effect sizes for communication quality ($d=-2.94$ and $d=-2.02$) confirm the quality-safety trade-off inherent in epistemic constraint application, discussed further in the discussion.

Cross-Seed Reproducibility

To assess reproducibility, we examined performance convergence across random seeds for the context-seeking metric, our primary measure of curiosity (figure 3). For GPT-4.1-mini, all five seeds showed consistent context-seeking rates under the BODHI condition, with tight clustering of seed-level estimates ranging from 92.9% to 100.0%, with three seeds achieving perfect 100% context-seeking behaviour. For GPT-4o-mini, all five seeds demonstrated similar patterns, with context-seeking rates between 68.7% and 78.3%. The shaded CIs overlap substantially across seeds, indicating that observed improvements in curiosity behaviour are reproducible rather than artefacts of particular random samples. Minor seed-to-seed variance was observed in context-seeking rates, reflecting sampling heterogeneity inherent in stratified vignette selection and stochastic text generation with temperature 0.7. Importantly, this variance did not alter the direction

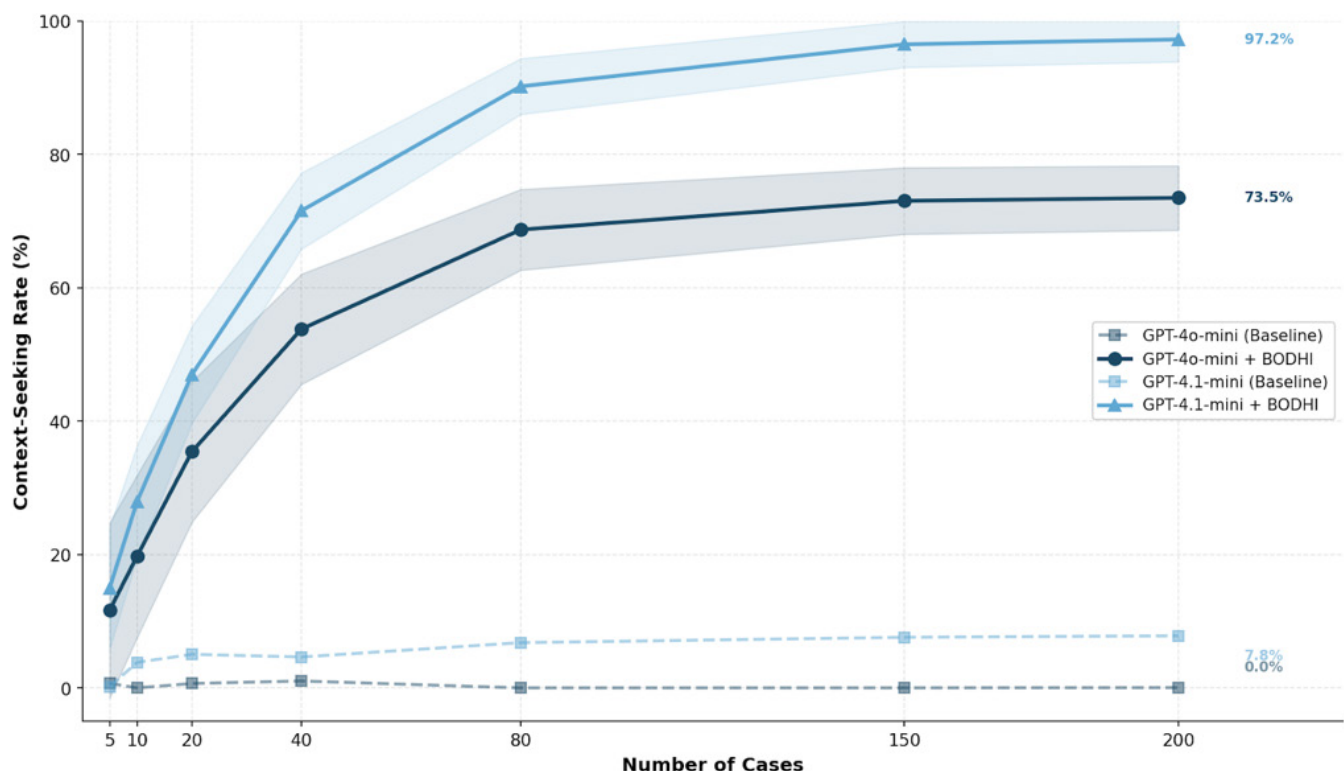


Figure 3 Cross-seed convergence of context-seeking rates for GPT-4.1-mini and GPT-4o-mini. Dashed lines represent baseline performance; solid lines represent BODHI performance. Shaded regions indicate $\pm$ SD across five seeds. Each line represents context-seeking performance across 200 vignettes. Consistent performance across seeds demonstrates reproducibility of the BODHI framework's curiosity-enhancing effects. Minor seed-to-seed variance reflects sampling heterogeneity and stochastic generation, not framework instability. BODHI, Balanced, Open-minded, Diagnostic, Humble and Inquisitive.

or statistical significance of effects. The consistent context-seeking performance across 5 independent replications with 200 vignettes each provides strong evidence for framework stability in eliciting curiosity-driven behaviour. Detailed seed-level results for all metrics are provided in online supplemental tables S5 and S6.

Quality-Safety Tradeoff

Communication quality scores decreased in both models (GPT-4.1-mini: -12.5pp, $d=-2.94$; GPT-4o-mini: -11.3pp, $d=-2.02$). This trade-off reflects the fundamental shift from confident declarative responses to more nuanced, question-containing outputs. The HealthBench rubric may undervalue epistemic humility, penalising responses that explicitly acknowledge uncertainty or request clarification. We argue this trade-off is appropriate for high-stakes clinical domains where overconfident communication poses greater risks than appropriately hedged responses. A detailed analysis of quality-safety trade-offs is provided in online supplemental material S4.3.

DISCUSSION

This study demonstrates that epistemic virtues can be operationalised in clinical AI through structured chain-of-thought prompting. BODHI achieved substantial improvements in context-seeking behaviour (+89.6pp for GPT-4.1-mini) and overall clinical quality (+16.6pp), with very large effect sizes suggesting robust behavioural modification rather than marginal improvement. Crucially, these gains were achieved through Chain of Thought alone, without model fine-tuning or architectural modifications, enabling immediate deployment across clinical settings.

BODHI differs fundamentally from existing approaches. Uncertainty quantification methods^{9–11} estimate confidence but do not modify how systems communicate or behave when uncertain. Sample consistency and token-level probability approaches^{7 22} achieve discrimination but with poor calibration and persistent overconfidence. Fine-tuning approaches^{15 23} require model modification and may not generalise across clinical domains. Conceptual frameworks calling for epistemic humility^{16 17} identify the problem without providing engineering solutions. In contrast, BODHI operates at the prompting level, requires no model modification and empirically demonstrates behavioural change through very large effect sizes on both curiosity ($d=16.38$ to 19.54) and humility ($d=5.80$ for GPT-4.1-mini) metrics.

The differential response between models merits consideration. GPT-4.1-mini showed stronger overall improvement (+16.6pp vs +2.2pp), suggesting that model capacity influences the effectiveness of epistemic constraint application. GPT-4o-mini achieved comparable context-seeking rates (73.5%) but lower overall scores, potentially reflecting differences in baseline reasoning capabilities or instruction-following fidelity. Both models showed substantial improvement in the primary epistemic

metrics, indicating that the two-pass protocol is effective across model variants, though optimisation may be required for specific deployment contexts.

The communication quality trade-off warrants careful interpretation. In high-stakes clinical domains, appropriately hedged, question-containing responses represent safer behaviour than confident declarations that may be incorrect. The observed decrease in communication quality scores may reflect rubric limitations rather than genuine degradation in clinical utility. Future evaluation frameworks should explicitly reward appropriate uncertainty expression and penalise overconfidence in safety-critical contexts, aligning assessment criteria with the epistemic virtues that clinical AI should embody.

Our findings suggest a potentially meaningful clinical impact, as BODHI does not merely estimate uncertainty but translates it into observable behaviour: By increasing context seeking, BODHI-equipped responses prompt clinicians to gather additional diagnostic information before committing to a clinical pathway, potentially reducing premature diagnostic closure. The substantial increase in hedging behaviour encourages shared decision making by presenting clinical reasoning as probabilistic rather than deterministic, enabling clinicians to weigh AI suggestions against their own expertise and patient-specific factors. Furthermore, the framework's escalation logic for high complexity, low confidence scenarios provides a structured safeguard against automation bias, ensuring that the most challenging cases are explicitly flagged for human judgement rather than resolved through overconfident model output. As AI decision support systems become increasingly embedded in clinical workflows, ensuring that these tools express appropriate uncertainty rather than unfounded confidence will hence be critical to reducing diagnostic error and preventing avoidable patient harm.

Clinical deployment requires integration with institutional governance structures. The framework interfaces at four key points: quarterly threshold calibration review by clinical informatics committees adjusting confidence and complexity boundaries based on outcome data, audit trail requirements logging all pass 1 analyses for retrospective review, structured override documentation when clinicians proceed against recommendations, and feedback loop closure linking patient outcomes to system recommendations for continuous improvement. A worked example demonstrating the complete BODHI workflow for emergency department chest pain triage is provided in online supplemental materials S5.3.

Limitations of this study include reliance on a single benchmark (HealthBench Hard), evaluation of two model families from one provider, and absence of clinician-in-the-loop validation. The two-pass architecture increases computational cost and latency, potentially limiting real-time applications. The framework's effectiveness may vary across clinical domains, patient populations

and institutional contexts. Additionally, while chain-of-thought explanations improve transparency, they may not fully reflect actual model computations, representing a limitation of post hoc rationalisation approaches. Future work should validate BODHI in prospective clinical settings with diverse patient populations and measure downstream outcomes including diagnostic accuracy and patient safety metrics.

CONCLUSIONS

BODHI provides an engineering framework for embedding curiosity and humility in clinical decision support AI. Unlike approaches that quantify uncertainty without behavioural modification or discuss epistemic virtues without operationalisation, BODHI's two-pass chain-of-thought protocol successfully constrains LLM behaviour to align with epistemic virtues. The dramatic improvements in context-seeking (+89.6pp) and substantial increases in hedging (+20.3pp) behaviour, combined with improved overall clinical quality (+16.6pp) and very large effect sizes across multiple metrics, demonstrate that LLMs can be reliably constrained to operate within epistemic boundaries when provided with structured prompting scaffolding. These findings suggest a pathway toward safer AI deployment where systems operate as collaborative partners that know when to ask questions and when to defer, rather than overconfident oracles that mask uncertainty behind declarative confidence.

The BODHI framework is available as an open-source Python package²⁴ with evaluation scripts at.²⁵

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